

Fourth Section.
The Elliptic Functions.

§ 46. The Weierstrass $\wp$ -function.

The linear transformations of the elliptic functions suggest that one should seek an elliptic function which, like the σ -function itself, is unchanged with respect to the linear transformations. In order to form such a function, we first put the equations (1) of the preceding paragraph into the following form:

$$\begin{aligned}\frac{\sigma_{10}(u)}{\sigma(u)} &= \frac{2K}{\omega_1} \frac{\operatorname{cn} v}{\operatorname{sn} v}, \\ \frac{\sigma_{00}(u)}{\sigma(u)} &= \frac{2K}{\omega_1} \frac{\operatorname{dn} v}{\operatorname{sn} v}, \\ \frac{\sigma_{01}(u)}{\sigma(u)} &= \frac{2K}{\omega_1} \frac{1}{\operatorname{sn} v},\end{aligned}\tag{1}$$

where, for abbreviation,

$$v = \frac{2Ku}{\omega_1}\tag{2}$$

is put. From this, with the aid of the relations

$$\begin{aligned}\operatorname{cn}^2 v &= 1 - \operatorname{sn}^2 v, & \operatorname{dn}^2 v &= 1 - \kappa^2 \operatorname{sn}^2 v, \\ \frac{\sigma_{01}^2(u)}{\sigma^2(u)} - \frac{\sigma_{10}^2(u)}{\sigma^2(u)} &= \frac{4K^2}{\omega_1^2}, & \frac{\sigma_{01}^2(u)}{\sigma^2(u)} - \frac{\sigma_{00}^2(u)}{\sigma^2(u)} &= \frac{4K^2}{\omega_1^2} \kappa^2,\end{aligned}$$

one concludes that three quantities e_1, e_2, e_3 , independent of u and therefore depending only on ω_1, ω_2 , can be so chosen that

$$\frac{\sigma_{10}^2(u)}{\sigma^2(u)} + e_1 = \frac{\sigma_{00}^2(u)}{\sigma^2(u)} + e_2 = \frac{\sigma_{01}^2(u)}{\sigma^2(u)} + e_3 = \wp(u),\tag{3}$$

where $\wp(u)$ is a doubly periodic function newly defined by (3). The quantities e_1, e_2, e_3 may be arbitrary, provided only that they satisfy the conditions

$$e_1 - e_3 = \frac{4K^2}{\omega_1^2}, \quad e_2 - e_3 = \frac{4K^2}{\omega_1^2} \kappa^2.\tag{4}$$

It is therefore still open to us, for their complete determination, to add the condition

$$e_1 + e_2 + e_3 = 0.\tag{5}$$

Then the following expressions for e_1, e_2, e_3 result:

$$\begin{aligned}e_1 &= \frac{4K^2}{\omega_1^2} \frac{1 + \kappa'^2}{3}, \\ e_2 &= -\frac{4K^2}{\omega_1^2} \frac{\kappa'^2 - \kappa^2}{3}, \\ e_3 &= -\frac{4K^2}{\omega_1^2} \frac{1 + \kappa^2}{3}.\end{aligned}\tag{6}$$

Consequently the function $\wp(u)$, which has the periods ω_1, ω_2 , is expressed by (1) as

$$\wp(u) = \frac{4K^2}{\omega_1^2} \left(\frac{1}{\operatorname{sn}^2 v} - \frac{1 + \kappa^2}{3} \right).\tag{7}$$

The function $\wp(u)$ has the desired property of invariance under linear transformations, as is seen from the expression

$$\wp(u) = \frac{1}{3} \frac{\sigma_{00}^2(u) + \sigma_{01}^2(u) + \sigma_{10}^2(u)}{\sigma^2(u)}. \quad (8)$$

In consequence of (3), the e_1, e_2, e_3 are interchanged under a linear transformation in the same way as the $\sigma_{10}, \sigma_{00}, \sigma_{01}$; a symmetric function of them is therefore unchanged under a linear transformation and is called an invariant. There are two fundamental invariants; we denote them by g_2, g_3 :

$$g_2 = -4(e_2e_3 + e_3e_1 + e_1e_2) = 2(e_1^2 + e_2^2 + e_3^2) = \frac{64}{3} \frac{K^4}{\omega_1^4} (1 - \kappa^2 \kappa'^2), \quad (9)$$

$$g_3 = 4e_1e_2e_3 = \frac{2^8}{27} \frac{K^6}{\omega_1^6} (2 + \kappa^2 \kappa'^2)(\kappa'^2 - \kappa^2). \quad (10)$$

We also adjoin the function known by the name of the discriminant:

$$G = (e_2 - e_3)^2(e_3 - e_1)^2(e_1 - e_2)^2 = \frac{1}{16} (g_2^3 - 27g_3^2) = \frac{(2K)^{12}}{\omega_1^{12}} \kappa^4 \kappa'^4, \quad (11)$$

for which, by § 43, (3), (15) and § 34, (2), one can also put

$$G = \frac{2^8 \pi^{12} \eta(\omega)^{24}}{\omega_1^{12}}. \quad (12)$$

$\wp(u), g_2, g_3, G$ are homogeneous functions, as the following relations show, where λ is an arbitrary factor:

$$\begin{aligned} \wp(\lambda u, \lambda \omega_1, \lambda \omega_2) &= \lambda^{-2} \wp(u, \omega_1, \omega_2), \\ \wp'(\lambda u, \lambda \omega_1, \lambda \omega_2) &= \lambda^{-3} \wp'(u, \omega_1, \omega_2), \\ g_2(\lambda \omega_1, \lambda \omega_2) &= \lambda^{-4} g_2(\omega_1, \omega_2), \\ g_3(\lambda \omega_1, \lambda \omega_2) &= \lambda^{-6} g_3(\omega_1, \omega_2), \\ G(\lambda \omega_1, \lambda \omega_2) &= \lambda^{-12} G(\omega_1, \omega_2). \end{aligned} \quad (13)$$

If, as in § 45, (2), we put

$$\omega_1 = 2K, \quad \omega_2 = 2iK',$$

then

$$\begin{aligned} e_1 &= \frac{1 + \kappa'^2}{3}, & e_2 &= -\frac{\kappa'^2 - \kappa^2}{3}, & e_3 &= -\frac{1 + \kappa^2}{3}, \\ g_2 &= \frac{4}{3} (1 - \kappa^2 \kappa'^2), & g_3 &= \frac{4}{27} (\kappa'^2 - \kappa^2)(2 + \kappa^2 \kappa'^2), \\ G &= \kappa^4 \kappa'^4. \end{aligned} \quad (14)$$

If we eliminate ω_1 from the general expressions (9), (10), (11) for g_2, g_3, G , we obtain homogeneous functions of order zero, hence functions of ω alone, which remain unchanged under linear transformations. Such functions are $g_2^3 : G, g_3^2 : G$. Among these functions, all of which can be reduced to one another, we single out one, which we denote by $j(\omega)$ and simply call the invariant, and define it thus:

$$j(\omega) = 2^8 \frac{(1 - \kappa^2 \kappa'^2)^3}{\kappa^4 \kappa'^4}, \quad (15)$$

from which we get

$$\frac{4 \cdot 27 g_2^3}{G} = j(\omega), \quad \frac{4 \cdot 27 \cdot 27 g_3^2}{G} = j(\omega) - 27 \cdot 64 = \frac{64(2 + \kappa^2 \kappa'^2)^2 (\kappa'^2 - \kappa^2)^2}{\kappa^4 \kappa'^4}. \quad (16)$$

Considered as a function of κ^2 , the function $j(\omega)$ has the property of remaining unchanged when κ^2 is replaced by one of the six values

$$\kappa^2, \quad \kappa'^2, \quad \frac{1}{\kappa^2}, \quad \frac{1}{\kappa'^2}, \quad -\frac{\kappa^2}{\kappa'^2}, \quad -\frac{\kappa'^2}{\kappa^2}.$$

If, according to (7), we form the differential quotient of the function $\wp(u)$, we obtain

$$\wp'(u) = -\frac{16K^3}{\omega_1^3} \frac{\operatorname{cn} v \operatorname{dn} v}{\operatorname{sn}^3 v} = -2 \frac{\sigma_{00}(u)\sigma_{10}(u)\sigma_{01}(u)}{\sigma(u)^3}, \quad (17)$$

and from this, by (3),

$$\begin{aligned} \wp'(u)^2 &= 4[\wp(u) - e_1][\wp(u) - e_2][\wp(u) - e_3] \\ &= 4\wp(u)^3 - g_2\wp(u) - g_3, \end{aligned} \quad (18)$$

or finally

$$du = \frac{d\wp}{\sqrt{4\wp^3 - g_2\wp - g_3}}, \quad (19)$$

from which one sees that the function $\wp(u)$ stands in the same relation to the Weierstrass normal form of the elliptic differential as the function $\operatorname{sn} v$ stands to the Legendre form.

§ 47. The elliptic transcendents of the second kind.

Jacobi introduced as transcendent of the second kind the function

$$Z(v) = \frac{\Theta'(v)}{\Theta(v)} = \frac{d \log \Theta(v)}{dv}. \quad (1)$$

The relation of this function to the elliptic integrals of the second kind results from formula (16) of § 23; it should immediately be observed that entirely similar considerations could be attached to the formulae (15), (17), (18) there, but that these would not lead to essentially new results.

If there one puts $v : 2K$ in place of v , it follows from § 42 and § 43 that

$$\begin{aligned} \frac{d^2 \log \Theta(v)}{dv^2} &= \frac{1}{4K^2} \frac{\vartheta_{01}''}{\vartheta_{01}} - \kappa^2 \operatorname{sn}^2 v \\ &= \frac{1}{4K^2} \frac{\vartheta_{01}''}{\vartheta_{01}} - 1 + \operatorname{dn}^2 v. \end{aligned} \quad (2)$$

We put

$$1 - \frac{\vartheta_{01}''}{4K^2 \vartheta_{01}} = \frac{E}{K}, \quad (3)$$

and obtain by integration of (2)

$$\frac{d \log \Theta(v)}{dv} = Z(v) = -\frac{E}{K}v + \int_0^v \operatorname{dn}^2 v \, dv,$$

or, putting

$$E(v) = \int_0^v \operatorname{dn}^2 v \, dv, \quad (4)$$

$$Z(v) = E(v) - \frac{E}{K}v, \quad \frac{dZ(v)}{dv} = \operatorname{dn}^2 v - \frac{E}{K}. \quad (5)$$

The functions $\Theta(v)$ and $\Theta(v + K)$ are even functions, and therefore $\Theta'(0)$ and $\Theta'(K)$ are equal to zero. Hence, if in (5) we put $v = K$, it follows that

$$E = \int_0^K \operatorname{dn}^2 v \, dv. \quad (6)$$

If one also introduces for dv the differential

$$\frac{1}{2} \frac{d\xi}{\sqrt{\xi(1-\xi)(1-\kappa^2\xi)}}$$

[§ 42, (8)], then one obtains for $E(v)$ an elliptic integral of the second kind (§ 11):

$$E(v) = \frac{1}{2} \int_0^\xi \frac{d\xi(1 - \kappa^2 \xi)}{\sqrt{\xi(1 - \xi)(1 - \kappa^2 \xi)}}, \quad (7)$$

$$E = \frac{1}{2} \int_0^1 \frac{d\xi(1 - \kappa^2 \xi)}{\sqrt{\xi(1 - \xi)(1 - \kappa^2 \xi)}}, \quad (8)$$

where the latter integral is to be taken in the same sense as in § 42, (17).

$E(v)$ and $Z(v)$ are odd functions of the argument. From the addition theorem of the Θ -function [§ 43, (11)] one obtains, by logarithmic differentiation,

$$Z(u + v) + Z(u - v) = 2Z(u) - \frac{2\kappa^2 \operatorname{sn}^2 v \operatorname{sn} u \operatorname{cn} u \operatorname{dn} u}{1 - \kappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}, \quad (9)$$

$$Z(u + v) - Z(u - v) = 2Z(v) - \frac{2\kappa^2 \operatorname{sn}^2 u \operatorname{sn} v \operatorname{cn} v \operatorname{dn} v}{1 - \kappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v}. \quad (10)$$

Here Z can also be replaced by E , and by addition one obtains [§ 44, (16)]

$$E(u) + E(v) - E(u + v) = \kappa^2 \operatorname{sn} u \operatorname{sn} v \operatorname{sn}(u + v). \quad (11)$$

From this one obtains the periodicity properties of the function $E(u)$ if one puts $v = \pm K$, $K + iK'$. But one also arrives at them in the following way.

From the first equation § 43, (3), it follows that

$$\frac{d \log H(v)}{dv} = Z(v + iK') + \frac{i\pi}{2K},$$

and hence from § 43, (6) and § 42, (11),

$$\begin{aligned} Z(v + iK') &= Z(v) + \frac{\operatorname{cn} v \operatorname{dn} v}{\operatorname{sn} v} - \frac{i\pi}{2K}, \\ Z(v + K) &= Z(v) - \frac{\kappa^2 \operatorname{sn} v \operatorname{cn} v}{\operatorname{dn} v}, \end{aligned} \quad (12)$$

and, if in the first of these formulae one puts $v = K$,

$$Z(K + iK') = -\frac{i\pi}{2K}. \quad (13)$$

By applying the formulae (12) twice [taking account of § 44, (18), (20)],

$$Z(v + 2iK') = Z(v) - \frac{i\pi}{K}, \quad Z(v + 2K) = Z(v). \quad (14)$$

Transferred to the function $E(v)$, these equations give

$$\begin{aligned} E(v + iK') &= E(v) + \frac{\operatorname{cn} v \operatorname{dn} v}{\operatorname{sn} v} + \frac{iEK'}{K} - \frac{i\pi}{2K}, \\ E(v + K) &= E(v) + E - \frac{\kappa^2 \operatorname{sn} v \operatorname{cn} v}{\operatorname{dn} v}, \end{aligned} \quad (15)$$

$$E(v + 2iK') = E(v) + \frac{2iEK'}{K} - \frac{i\pi}{K}, \quad E(v + 2K) = E(v) + 2E. \quad (16)$$

If in (15) we put $v = K$ and define a quantity E' by the equation

$$E(K + iK') = E' + i(K' - E'), \quad (17)$$

then one obtains the formula known under the name of Legendre's relation,

$$EK' + KE' - KK' = \frac{\pi}{2}. \quad (18)$$

The meaning of the quantity E' introduced here is recognized if one derives Legendre's relation by a second path, using one of the linear transformations. It results, if in § 31, (11), one puts

$$u = \frac{v}{2K}, \quad \omega = \frac{iK'}{K}, \quad \frac{u}{\omega} = \frac{v}{2iK'},$$

that, by § 43, (1), (4),

$$\sqrt{K} e^{-\frac{\pi v^2}{4KK'}} \Theta(iv, \kappa') = \sqrt{K'} H(v + K), \quad (19)$$

and from this, by logarithmic differentiation,

$$iZ(iv, \kappa') = \frac{\pi v}{2KK'} + \frac{H'(v + K)}{H(v + K)}, \quad (20)$$

or, by § 43, (6),

$$iZ(iv, \kappa') = \frac{\pi v}{2KK'} + Z(v) + \frac{d \log \operatorname{cn} v}{dv}. \quad (21)$$

But by (5)

$$i \frac{dZ(iv, \kappa')}{dv} = -\operatorname{dn}^2(iv, \kappa') + \frac{E'}{K'},$$

if E' now has the meaning

$$E' = \int_0^{K'} \operatorname{dn}^2(v, \kappa') dv, \quad (22)$$

that is, arises from E by interchanging κ with κ' . If, however, in (21) one puts $v = 0$ after having first differentiated, then the relation (18) again results; from this it follows that E' is in both cases the same quantity.

§ 48. The elliptic transcendents of the third kind.

If we integrate the formula (10) of the preceding paragraph,

$$\frac{\Theta'(u+v)}{\Theta(u+v)} - \frac{\Theta'(u-v)}{\Theta(u-v)} = 2Z(v) - \frac{2\kappa^2 \operatorname{sn}^2 u \operatorname{sn} v \operatorname{cn} v \operatorname{dn} v}{1 - \kappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v},$$

with respect to u , we get

$$\frac{1}{2} \log \frac{\Theta(u-v)}{\Theta(u+v)} + uZ(v) = \int_0^u \frac{\kappa^2 \operatorname{sn}^2 u \operatorname{sn} v \operatorname{cn} v \operatorname{dn} v}{1 - \kappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v} du, \quad (1)$$

and we can also add the formula, to be derived in the same way [from § 43, (11)],

$$\frac{1}{2} \log \frac{H(v-u)}{H(v+u)} + uZ(v) = \int_0^u \frac{\operatorname{sn} v \operatorname{cn} v \operatorname{dn} v du}{\operatorname{sn}^2 u - \operatorname{sn}^2 v}, \quad (2)$$

which, moreover, can also be derived from (1). We now put, with Jacobi,

$$\Pi(u, v) = \int_0^u \frac{\kappa^2 \operatorname{sn}^2 u \operatorname{sn} v \operatorname{cn} v \operatorname{dn} v}{1 - \kappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v} du, \quad (3)$$

and call $\Pi(u, v)$ the transcendent of the third kind with argument u and parameter v . If one replaces du by the expression

$$\frac{d\xi}{2\sqrt{\xi(1-\xi)(1-\kappa^2\xi)}}$$

and $\operatorname{sn}^2 u$ by ξ , then both (1) and (2) give an elliptic integral of the third kind, as we have learned it in § 11.

From (1) follows first

$$\Pi(u, v) - uZ(v) = \Pi(v, u) - vZ(u), \quad (4)$$

or Jacobi's theorem on the interchange of argument and parameter. The function $\Pi(u, v)$ vanishes when $v = K$ or $v = K + iK'$, because for the former value $\operatorname{cn} v$, and for the latter $\operatorname{dn} v$, is equal to zero. If therefore we insert these values for u in (4), it follows that

$$\begin{aligned} \Pi(K, v) &= KZ(v), \\ \Pi(K + iK', v) &= (K + iK')Z(v) + \frac{i\pi v}{2K} \quad [\S 47, (13)], \end{aligned} \quad (5)$$

whereby the complete integrals of the third kind are led back to the second kind.

We still wish to derive the addition theorem for the Π -function, which Jacobi gives in the *Fundamenta nova*, art. 53–55, in various forms that can be reduced to one another only with difficulty. First, from the definition (1), if we now denote the parameter by a , we obtain

$$\begin{aligned} \Pi(u + v, a) - \Pi(u, a) - \Pi(v, a) \\ = \frac{1}{2} \log \frac{\Theta(u + v - a)\Theta(u + a)\Theta(v + a)}{\Theta(u + v + a)\Theta(u - a)\Theta(v - a)}, \end{aligned} \quad (6)$$

and it remains only to represent the expression standing under the logarithm by elliptic functions.

This is first done easily, as at the cited place of the *Fundamenta*, by means of the formula [§ 43, (11)]

$$\Theta^2(0)\Theta(u + v)\Theta(u - v) = \Theta^2(u)\Theta^2(v)[1 - \kappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v],$$

if therein one puts, for u, v , first $u \pm a, v \pm a$ and then $\pm a, u + v \pm a$. One obtains thus

$$\begin{aligned} \Theta^2(0)\Theta(u + v \pm 2a)\Theta(u - v) &= \Theta^2(u \pm a)\Theta^2(v \pm a)[1 - \kappa^2 \operatorname{sn}^2(u \pm a) \operatorname{sn}^2(v \pm a)], \\ \Theta^2(0)\Theta(u + v \pm 2a)\Theta(u + v) &= \Theta^2(a)\Theta^2(u + v \pm a)[1 - \kappa^2 \operatorname{sn}^2 a \operatorname{sn}^2(u + v \pm a)], \end{aligned}$$

from which follows

$$\frac{\Theta(u + v - a)\Theta(u + a)\Theta(v + a)}{\Theta(u + v + a)\Theta(u - a)\Theta(v - a)} = \sqrt{\frac{[1 - \kappa^2 \operatorname{sn}^2(u - a) \operatorname{sn}^2(v - a)][1 - \kappa^2 \operatorname{sn}^2 a \operatorname{sn}^2(u + v + a)]}{[1 - \kappa^2 \operatorname{sn}^2(u + a) \operatorname{sn}^2(v + a)][1 - \kappa^2 \operatorname{sn}^2 a \operatorname{sn}^2(u + v - a)]}}. \quad (7)$$

A second expression is obtained from the addition formula (13), § 22, which can be represented in the form

$$\Theta(0)\Theta(u \pm a)\Theta(v \pm a)\Theta(u + v) = \Theta(u)\Theta(v)\Theta(a)\Theta(u + v \pm a)[1 \pm \kappa^2 \operatorname{sn} a \operatorname{sn} u \operatorname{sn} v \operatorname{sn}(u + v \pm a)],$$

and hence, by division,

$$\frac{\Theta(u + v - a)\Theta(u + a)\Theta(v + a)}{\Theta(u + v + a)\Theta(u - a)\Theta(v - a)} = \frac{1 + \kappa^2 \operatorname{sn} a \operatorname{sn} u \operatorname{sn} v \operatorname{sn}(u + v + a)}{1 - \kappa^2 \operatorname{sn} a \operatorname{sn} u \operatorname{sn} v \operatorname{sn}(u + v - a)}. \quad (8)$$

Jacobi gives still a third expression for the Θ -quotients in (6):

$$\frac{\{1 - \kappa^2 \operatorname{sn}^2(\frac{u-v}{2}) \operatorname{sn}^2(\frac{u+v}{2} + a)\} \{1 - \kappa^2 \operatorname{sn}^2(\frac{u+v}{2}) \operatorname{sn}^2(\frac{u+v}{2} - a)\}}{\{1 - \kappa^2 \operatorname{sn}^2(\frac{u-v}{2}) \operatorname{sn}^2(\frac{u+v}{2} - a)\} \{1 - \kappa^2 \operatorname{sn}^2(\frac{u+v}{2}) \operatorname{sn}^2(\frac{u+v}{2} + a)\}}. \quad (9)$$

The direct conversion of the three expressions (7), (8), (9) into one another succeeds most simply if one starts from the formula last given by Jacobi,

$$1 - \kappa^2 \operatorname{sn}(a + u) \operatorname{sn}(a - u) \operatorname{sn}(a + v) \operatorname{sn}(a - v) = \frac{(1 - \kappa^2 \operatorname{sn}^4 a)(1 - \kappa^2 \operatorname{sn}^2 u \operatorname{sn}^2 v)}{(1 - \kappa^2 \operatorname{sn}^2 a \operatorname{sn}^2 u)(1 - \kappa^2 \operatorname{sn}^2 a \operatorname{sn}^2 v)}. \quad (10)$$

The verification of this formula is easy because, with the aid of § 44, (13), the right and left sides become rational functions of $\operatorname{sn}^2 u, \operatorname{sn}^2 v$, the identity of which is immediately evident.

If in this formula one replaces

$$u, \quad v, \quad a$$

by

$$\frac{u-v}{2}, \quad \frac{u+v}{2} \pm a, \quad \frac{u+v}{2},$$

one obtains

$$\begin{aligned} & 1 \pm \kappa^2 \operatorname{sn} u \operatorname{sn} v \operatorname{sn} a \operatorname{sn}(u+v \pm a) \\ &= \frac{[1 - \kappa^2 \operatorname{sn}^4(\frac{u+v}{2})][1 - \kappa^2 \operatorname{sn}^2(\frac{u-v}{2}) \operatorname{sn}^2(\frac{u+v}{2} \pm a)]}{[1 - \kappa^2 \operatorname{sn}^2(\frac{u+v}{2}) \operatorname{sn}^2(\frac{u-v}{2})][1 - \kappa^2 \operatorname{sn}^2(\frac{u+v}{2}) \operatorname{sn}^2(\frac{u+v}{2} \pm a)]}, \end{aligned}$$

and, if the two formulae contained herein are divided by one another, the agreement of (8) and (9) results.

Putting $v = u$ in (10) gives

$$1 - \kappa^2 \operatorname{sn}^2(a+u) \operatorname{sn}^2(a-u) = \frac{(1 - \kappa^2 \operatorname{sn}^4 a)(1 - \kappa^2 \operatorname{sn}^4 u)}{(1 - \kappa^2 \operatorname{sn}^2 a \operatorname{sn}^2 u)^2}. \quad (11)$$

If herein one replaces first

$$a, \quad u$$

by

$$\frac{u+v}{2} \pm a, \quad -\frac{u+v}{2},$$

and then by

$$\frac{u+v}{2} \pm a, \quad \frac{u-v}{2},$$

four formulae result:

$$\begin{aligned} 1 - \kappa^2 \operatorname{sn}^2 a \operatorname{sn}^2(u+v \pm a) &= \frac{[1 - \kappa^2 \operatorname{sn}^4(\frac{u+v}{2} \pm a)](1 - \kappa^2 \operatorname{sn}^4 \frac{u+v}{2})}{[1 - \kappa^2 \operatorname{sn}^2(\frac{u+v}{2} \pm a) \operatorname{sn}^2 \frac{u+v}{2}]^2}, \\ 1 - \kappa^2 \operatorname{sn}^2(u \pm a) \operatorname{sn}^2(v \pm a) &= \frac{[1 - \kappa^2 \operatorname{sn}^4(\frac{u+v}{2} \pm a)](1 - \kappa^2 \operatorname{sn}^4 \frac{u-v}{2})}{[1 - \kappa^2 \operatorname{sn}^2(\frac{u+v}{2} \pm a) \operatorname{sn}^2 \frac{u-v}{2}]^2}, \end{aligned}$$

from which the agreement of (9) with (7) results.

§ 49. The transcendents of the second and third kind of Weierstrass.

It remains to set up the transcendents of the second and third kind in the Weierstrass form. We reach the former similarly as in § 47, by twice differentiating $\log \sigma(u)$. Thus, from

$$\sigma(u) = \omega_1 e^{\eta_1 u^2 / \omega_1} \frac{\vartheta_{11}(u/\omega_1)}{\vartheta'_{11}} \quad [\S 37, (4)], \quad (1)$$

one obtains

$$\begin{aligned} \frac{d^2 \log \sigma(u)}{du^2} &= \frac{2\eta_1}{\omega_1} + \frac{d^2 \log \vartheta_{11}(u/\omega_1)}{du^2}, \\ &= \frac{2\eta_1}{\omega_1} + \frac{1}{\omega_1^2} \frac{\vartheta''_{00}}{\vartheta_{00}} - \frac{1}{\omega_1^2} \frac{\vartheta_{11}^2}{\vartheta_{00}^2} \frac{\vartheta_{00}^2(u/\omega_1)}{\vartheta_{11}^2(u/\omega_1)} \quad [\S 23, (18)], \end{aligned}$$

and therefore, if the ϑ -quotient is expressed by § 42 through elliptic functions, and these by § 46 through the $\wp$ -function,

$$\frac{d^2 \log \sigma(u)}{du^2} = A - \wp(u),$$

where A is a constant. This, however, is found if the equation is written according to § 46, (8), in the form

$$\sigma''(u)\sigma(u) - \sigma'(u)\sigma'(u) = A\sigma^2(u) - \frac{1}{3}[\sigma_{00}^2(u) + \sigma_{01}^2(u) + \sigma_{10}^2(u)].$$

If one differentiates this twice and then puts $u = 0$, there results $A = 0$ [§ 35, (12), § 37, (5), (7)], and we get

$$\frac{d^2 \log \sigma(u)}{du^2} = -\wp(u), \quad (2)$$

a formula which, for Weierstrass, serves for the definition of the $\wp$ -function.

If therefore

$$\frac{\sigma'(u)}{\sigma(u)} = \zeta(u) \quad (3)$$

is put, then $\zeta(u)$ is a single-valued function of u , and indeed an elliptic integral of the second kind:

$$\zeta(u) = - \int \wp(u) du = - \int \frac{\wp d\wp}{\sqrt{4\wp^3 - g_2\wp - g_3}} \quad [\S 46, (19)], \quad (4)$$

where the additive constant is determined by the requirement that $\zeta(u)$ be an odd function. The periodicity of the ζ -function results from the period equations of the σ -function [§ 35, (9)]:

$$\zeta(u + \omega_1) - \zeta(u) = 2\eta_1, \quad \zeta(u + \omega_2) - \zeta(u) = 2\eta_2. \quad (5)$$

Thus $2\eta_1$ and $2\eta_2$ are represented as complete integrals of the second kind, analogous to E, E' :

$$2\eta_1 = - \int_u^{u+\omega_1} \wp(u) du, \quad 2\eta_2 = - \int_u^{u+\omega_2} \wp(u) du, \quad (6)$$

with the equation corresponding to Legendre's relation

$$\eta_1\omega_2 - \eta_2\omega_1 = \pi i \quad [\S 35, (10)]. \quad (7)$$

To form the addition theorems for the Weierstrass transcendents, we start from the σ -function. For it, from (1), with the use of the addition formula for ϑ_{11} , § 22, (4), we obtain

$$\frac{\sigma(u+v)\sigma(u-v)}{\sigma^2(u)\sigma^2(v)} = \wp(v) - \wp(u), \quad (8)$$

and hence, by logarithmic differentiation,

$$\begin{aligned} \zeta(u+v) + \zeta(u-v) - 2\zeta(u) &= \frac{\wp'(u)}{\wp(u) - \wp(v)}, \\ \zeta(u+v) - \zeta(u-v) - 2\zeta(v) &= -\frac{\wp'(v)}{\wp(u) - \wp(v)}, \\ \zeta(u+v) &= \zeta(u) + \zeta(v) + \frac{1}{2} \frac{\wp'(u) - \wp'(v)}{\wp(u) - \wp(v)}. \end{aligned} \quad (9-10)$$

From this, by differentiating once more, one derives the addition theorem for the $\wp$ -function. Differentiating (9) with respect to u and v gives

$$\begin{aligned} \wp(u+v) + \wp(u-v) - 2\wp(u) &= -\frac{\wp''(u)}{\wp(u) - \wp(v)} + \frac{\wp'(u)^2}{[\wp(u) - \wp(v)]^2}, \\ 2\wp(u+v) - 2\wp(u-v) &= -\frac{2\wp'(u)\wp'(v)}{[\wp(u) - \wp(v)]^2}, \\ \wp(u+v) + \wp(u-v) - 2\wp(v) &= \frac{\wp''(v)}{\wp(u) - \wp(v)} + \frac{\wp'(v)^2}{[\wp(u) - \wp(v)]^2}, \end{aligned}$$

and, adding,

$$4\wp(u+v) - 2\wp(u) - 2\wp(v) = \left(\frac{\wp'(u) - \wp'(v)}{\wp(u) - \wp(v)} \right)^2 - \frac{\wp''(u) - \wp''(v)}{\wp(u) - \wp(v)}.$$

But

$$\begin{aligned} \wp'(u)^2 &= 4\wp(u)^3 - g_2\wp(u) - g_3, & \wp''(u) &= 6\wp(u)^2 - \frac{1}{2}g_2, \\ \wp''(u) - \wp''(v) &= 6[\wp^2(u) - \wp^2(v)], \end{aligned}$$

from which one obtains

$$\wp(u+v) + \wp(u) + \wp(v) = \frac{1}{4} \left(\frac{\wp'(u) - \wp'(v)}{\wp(u) - \wp(v)} \right)^2, \quad (11)$$

in agreement with formula (21), § 13.

Finally one obtains, by integration of the second equation (9) with respect to u ,

$$\frac{1}{2} \log \frac{\sigma(v-u)}{\sigma(v+u)} + u\zeta(v) = \frac{1}{2} \int_0^u \frac{\wp'(v)}{\wp(u) - \wp(v)} du, \quad (12)$$

whereby an integral of the third kind with argument u and parameter v is expressed by the σ -function. Another form of the integral of the third kind is obtained by integration of (10):

$$\log \frac{\sigma(u+v)}{\sigma(u)\sigma(v)} - u\zeta(v) = \frac{1}{2} \int \frac{\wp'(u) - \wp'(v)}{\wp(u) - \wp(v)} du. \quad (13)$$

§ 50. Expansions of the elliptic functions.

If in the expansions of the theta-quotients derived in § 26 and collected in Table II one puts $a = 0$, then, if one first omits the formulae which contain the factor $\vartheta_{11}(a)$ in the denominator, one obtains the partial-fraction expansions of twelve elliptic functions:

$$\begin{array}{ccc} \operatorname{sn} 2Ku, & \operatorname{cn} 2Ku, & \operatorname{dn} 2Ku, \\ \frac{1}{\operatorname{sn} 2Ku}, & \frac{\operatorname{cn} 2Ku}{\operatorname{sn} 2Ku}, & \frac{\operatorname{dn} 2Ku}{\operatorname{sn} 2Ku}, \\ \frac{1}{\operatorname{cn} 2Ku}, & \frac{\operatorname{sn} 2Ku}{\operatorname{cn} 2Ku}, & \frac{\operatorname{dn} 2Ku}{\operatorname{cn} 2Ku}, \\ \frac{1}{\operatorname{dn} 2Ku}, & \frac{\operatorname{sn} 2Ku}{\operatorname{dn} 2Ku}, & \frac{\operatorname{cn} 2Ku}{\operatorname{dn} 2Ku}. \end{array}$$

Thus, from formula (2), Table II, one obtains

$$\begin{aligned} \frac{\vartheta_{11}\vartheta_{10}(u)}{\pi\vartheta_{11}(u)\vartheta_{10}} &= \cotg \pi u - 2i \sum (-1)^{m/2} q^m \left(\frac{1}{e^{-2\pi i u} - q^m} - \frac{1}{e^{2\pi i u} - q^m} \right) \\ &= \cotg \pi u + 4 \sin 2\pi u \sum \frac{(-1)^{m/2} q^m}{1 - 2q^m \cos 2\pi u + q^{2m}}, \end{aligned}$$

where m runs through the sequence of positive even numbers $2, 4, 6, 8, \dots$. But by § 42, (4),

$$\frac{\vartheta_{10}(u)}{\vartheta_{11}(u)} = \frac{\vartheta_{00} \operatorname{cn} 2Ku}{\vartheta_{01} \operatorname{sn} 2Ku},$$

and from this it follows that

$$\frac{2K}{\pi} \frac{\operatorname{cn} 2Ku}{\operatorname{sn} 2Ku} = \cotg \pi u + 4 \sin 2\pi u \sum (-1)^{m/2} q^m \over 1 - 2q^m \cos 2\pi u + q^{2m}. \quad (1)$$

If the same procedure is applied in formula (2) of Table III, one obtains

$$\frac{2K \operatorname{cn} 2Ku}{\pi \operatorname{sn} 2Ku} = \cotg \pi u + 4 \sum (-1)^{m'/2} q^{mm'/2} \sin \pi m u.$$

Here m and m' run independently through the sequences of positive even numbers, and the summation with respect to m' can still be carried out:

$$\sum (-1)^{m'/2} q^{mm'/2} = -\frac{q^m}{1+q^m},$$

therefore

$$\frac{2K \operatorname{cn} 2Ku}{\pi \operatorname{sn} 2Ku} = \cotg \pi u - 4 \sum \frac{q^m}{1+q^m} \sin \pi m u. \quad (2)$$

While the first two formulae converge for all values of u , the convergence of the third is restricted to the region in which the imaginary part of u is, in absolute value, smaller than the imaginary part of ω . The formula is therefore valid for real u in any case.

In Table IV the expansions for the twelve functions are collected.

Among the sixteen formulae of Tables II and III there are four which contain the factor $\vartheta_{11}(a)$ in the denominator, so that one cannot at once put $a = 0$ in them. These are the formulae (1), (5), (9), (13). If one expands in these formulae according to ascending powers of a , the expansions on the right and on the left begin with a^{-1} ; equating on both sides the terms independent of a gives the required expansions.

One can proceed, for example, by changing a into $-a$ in formula (1) and taking the arithmetic mean of the two formulae. Then on the right the infinite term $1 : \cotg \pi a$ cancels out and on the left one obtains

$$\frac{\vartheta'_{11}[\vartheta_{11}(v+a) - \vartheta_{11}(v-a)]}{2\pi\vartheta_{11}(v)\vartheta_{11}(a)} = \frac{\vartheta''_{11}(v)}{\pi\vartheta_{11}(v)} \quad (a=0).$$

On the right side of (1) in Table II one obtains

$$\cotg \pi v - 4i \sum \left(\frac{q^m}{e^{-2\pi i v} - q^m} - \frac{q^m}{e^{2\pi i v} - q^m} \right) = \cotg \pi v + 4 \sum \frac{q^m \sin 2\pi v}{1 - 2q^m \cos 2\pi v + q^{2m}},$$

and on the right side of formula (1) of Table III one obtains

$$\cotg \pi v + 4 \sum \frac{q^{m/2}}{1 - q^m} \sin m\pi v.$$

If one also introduces Jacobi's H -function and writes u for v , one obtains

$$\frac{2K}{\pi} \frac{d \log H(2Ku)}{d(2Ku)} = \cotg \pi u + 4 \sin 2\pi u \sum \frac{q^m}{1 - 2q^m \cos 2\pi u + q^{2m}} = \cotg \pi u + 4 \sum \frac{q^m}{1 - q^m} \sin m\pi u. \quad (3)$$

On the left stands a transcendent of the second kind, which by § 47, (12), can be expressed through the Z -function:

$$\frac{2K}{\pi} Z(2Ku) + \frac{2K \operatorname{cn} 2Ku \operatorname{dn} 2Ku}{\pi \sin 2Ku},$$

for which one can also put

$$\frac{2K}{\pi} Z(2Ku) + \frac{d \log \sin 2Ku}{\pi du}.$$

In Table V the four formulae which result in this way are collected. If one subtracts formula (3) of this table from the three remaining ones, expansions result for the logarithmic derivatives of the three elliptic fundamental functions; of these we shall cite one:

$$\frac{d \log \operatorname{sn} 2Ku}{\pi du} = \frac{d \log \sin \pi u}{\pi du} + 4 \sin 2\pi u \sum_{h=1}^{\infty} \frac{(-1)^h q^h}{1 - 2q^h \cos 2\pi u + q^{2h}} = \frac{d \log \sin \pi u}{\pi du} - 4 \sum_{h=1}^{\infty} \frac{q^h \sin 2h\pi u}{1 + q^h}. \quad (4)$$

Expansions for the transcendents of the third kind result when one integrates the formulae of the fifth table between the limits $u + v$, $u - v$. Thus, for example, from formula (3) of this table one obtains

$$\frac{1}{2} \log \frac{\vartheta[2K(u-v)]}{\vartheta[2K(u+v)]} = -4 \sum \frac{q^{m/2}}{m(1-q^m)} \sin m\pi u \sin m\pi v. \quad (5)$$

Putting $u = 0$ in the formulae of Tables IV and V gives the following expansions:

$$\begin{aligned} \frac{2K}{\pi} &= 1 + 4 \sum \frac{q^{m/2}}{1+q^m} = 1 + 4 \sum \frac{(-1)^{(n-1)/2} q^n}{1-q^n}, \\ \frac{2\kappa K}{\pi} &= 4 \sum \frac{q^{n/2}}{1+q^n} = 4 \sum \frac{(-1)^{(n-1)/2} q^{n/2}}{1-q^n}, \\ \frac{2\kappa' K}{\pi} &= 1 + 4 \sum \frac{(-1)^{m/2} q^{m/2}}{1+q^m} = 1 - 4 \sum \frac{(-1)^{(n-1)/2} q^n}{1+q^n}. \end{aligned}$$

Finally one obtains an expansion for the complete integral of the second kind E from the formula

$$Z(v) = E(v) - \frac{E}{K}v.$$

If herein, after differentiation, one puts $v = 0$, it follows that

$$Z'(0) = \frac{K - E}{K},$$

and hence, by Table V, formula (3),

$$K - E = \frac{2\pi^2}{K} \sum \frac{q^n}{(1-q^n)^2} = \frac{\pi^2}{K} \sum \frac{mq^{m/2}}{1-q^m}.$$